

Attendance Code:
2750



4155

Week 1 - Introduction

Dr. James Rudd
jrudd@uncc.edu
Mobile: 704-965-1075



UNC CHARLOTTE

Our Teaching Assistants:

	Name(s)	Class	Email(s)	TA Hours
Vishnu Manoj  	Vishnu Vardhan Darimidi Manoj Kumar Reddy Janapala	ITSC 4155-001 10:15 – 1:00	vdarimid@uncc.edu mjanapal@uncc.edu	Vishnu: Wednesday - 11:00AM to 1:00PM Manoj: Thursday - 12:00PM to 2:00PM
Sai Sai  	Sai Kumar Reddy Bommareddy Sai Tharun Reddy Rupireddy	ITSC 4155-052 2:30 – 5:15	sbommar1@uncc.edu u srupired@uncc.edu	Sai: Thursday - 10:00AM to 12:00PM Tharun: Wednesday - 2:30PM to 4:30PM
	Srikar Chamarthy	ITSC 4155-090 5:30 – 8:15	schamar1@uncc.edu	Srikar: Tuesday - 11:00AM to 1:00PM

Canvas Link:

<https://uncc.instructure.com/courses/208588>

(Let us know if it doesn't work)



The goals of this class

- Build something of value while learning to work happily in a group setting
- Learn more about how things really work in the IT business without being left used, abused, and burnt out
- Get a good grade
- Get a great job
- Live happily ever after



Zoom Sign In Information (2:30 and 5:30)

<https://charlotte-edu.zoom.us/j/91542966452>



Zoom Decorum

- The standards set in place for on-line class participation:
 - To be available and responsive during scheduled class time.
 - Duties, obligations, and responsibilities are the same as with on-campus class, including the obligation to respond to questions, emails, assignments, projects, etc., in a timely manner.
 - To maintain an appropriate level of communication with the instructors, project team members and classmates.
 - Never connect to the class from a moving vehicle or in other unsafe situations. If noted by the instructor, you will be disconnected.
 - During class meetings, it is expected to have the camera "on" during class. You may set up virtual Charlotte background in place.
 - If you do not have a camera or are not willing to turn it on in these situations, **YOU SHOULD DROP THIS SECTION** and sign up for an on-campus section.
 - Maintain professionally acceptable standards of online behavior. Please maintain professional surroundings or use virtual Charlotte backgrounds.



Today, computer scientists are in the driver's seat

Case in point.....



Who makes more money? Raise your hand

1. Pro athlete
2. Software engineer





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**IF YOU WANT TO BE A
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A SOFTWARE DEVELOPER THAN A
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It's better to be a software developer than a pro athlete! Find out Why...

<https://www.businessinsider.com/better-to-be-a-software-engineer-than-a-pro-athlete-2017-1>



UNC CHARLOTTE

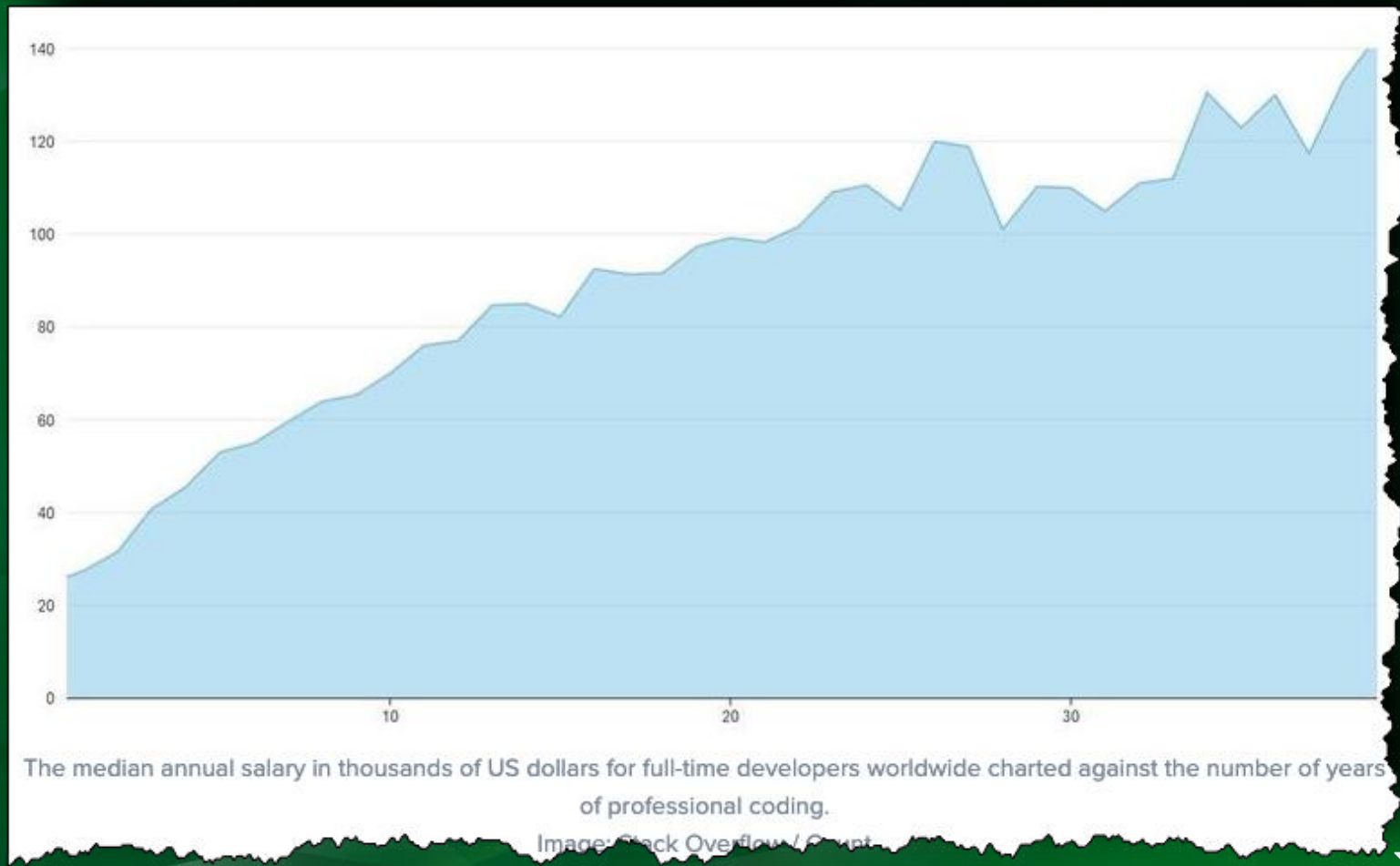
Good News Fridays

- There's no question being an elite pro athlete pays very, very well.
- NFL Players belong to a very exclusive club in America. It is possible for All-Pros LIKE Tom Brady and Aaron Rogers to make \$30M in a single year!
- But in reality, the average NFL career is just three years long and median career earnings for NFL players are about \$3 million.
- In the MLB, the average career length is 5.6 years with the lifetime earning potential of \$3,00,5180. None of that is chicken scratch. But if you factor in the number of people who try to become pro sports players and don't ever get a decent contract at all, the percentages would be far lower.
- Now, compare that to becoming a software engineer. Once you're out in the workforce, the average career span is 35 years with an average annual salary at \$125,418. The lifetime earnings potential adds up to \$5,016,723, better than both the typical pro NFL player and typical pro MLB athlete

<https://www.businessinsider.com/better-to-be-a-software-engineer-than-a-pro-athlete-2017-1>



You can make some good coin as an IT professional.. All the while having fun



Week 1 Agenda

Goals for Today

- Introductions
- Syllabus
- Week at a Glance - Assignments
- Grading
- Background Survey (used for assignment to groups)
- (6 per group max)

Goals for Week 2

- Software Development Process Preview
- Divide into your assigned groups / Acclimate with project members
- Learn what is required for project proposals
- Begin work on Project Proposals and Presentations



Syllabus



ITCS 4155
Software Development Projects
Syllabus

Professor: Dr. Jim Rudd
Office Hours: By appointment between the hours of 11am – 2pm Fridays
Phone: 704-965-1075
E-mail: jrudd@uncc.edu

1. Class Objectives

A software project succeeds when it meets the expectations of the people who use it over a long period of time.

The objective of the class is to explain software development as an engineering discipline. The course will review modern methods of software development, process models, software architecture, architectural styles, design patterns, assessment methodologies and tools.

These methods are to be applied in a comprehensive development project.

2. Expected Learning Outcomes

Upon successful completion of this course, students should have:

1. Ability to apply a disciplined approach to software requirements definition, detailed software design and implementation.
2. Ability to evaluate project economics and organize configuration management.
3. Ability to design and perform software quality assurance techniques, including testing and peer reviews.
4. As part of a team, apply software design knowledge to a significant software development project



3. Topics Covered

4155 Week by Week Glance

- ✓ Course Intro (Week 1)
- ✓ Ethics and Proposal Prep (Week 2)
- ✓ Agile Inception Presentations (Week 3)
- ✓ Communications, Intro to Agile, User Stories (Week 4)
- ✓ Prototyping (Week 5)
- ✓ Lab (Week 6)
- ✓ Sprint 1 Playback (Week 7)
- ✓ Estimating (Week 8)
- ✓ Velocity (Week 9)
- ✓ Sprint 2 Playback (Week 10)
- ✓ Architectural Patterns (Week 11)
- ✓ UML Diagramming (Week 12)
- ✓ Sprint 3 Due / No class (Week 13)
- ✓ Elevator Pitch and the Portfolio (Week 14)
- ✓ Sprint 4 Lab (Week 15)
- ✓ Sprint 4 Playback (Week 16)

Note: This is a capstone course in which you should be applying your existing knowledge of computer science to develop a software system. Having a working piece of software that meets the project requirements at the end of the course is emphasized. Every student will be expected to participate in design and implementation of a part of a large system.

You will not learn how to program in this course; you should already know how to program. If you are not comfortable with developing a large software system or work in



a group, there are other options available to you to fulfill the capstone experience in computer science. For example, another option is to work with a faculty member on a semester-long research project (ITCS 4160/4161).

4. Tests and Quizzes

There will not be a mid-term or final test. Instead quizzes will be used to provide clear and timely feedback on pupil's content understanding.

5. Project

A comprehensive team-based project is the cornerstone of this class. Class participants will organize into six-person project teams.

The projects will come from class and team discussions and proposed projects will require instructor approval. Project development will follow agile software lifecycle dynamics.

Formal deliverables will include requirements and design documents and multiple incremental versions of executable software, assessed for the level of completion through class demonstrations. The projects will follow a strict delivery schedule, discussed and defined in class. Each day of deliverables delay will cost you 5% of the grade.

In the event industry brings project ideas to the class, please note that your participation in an industry-sponsored project may expose you to proprietary information. You will be required to sign non-disclosure forms that address the proper handling of proprietary information.



Syllabus - continued

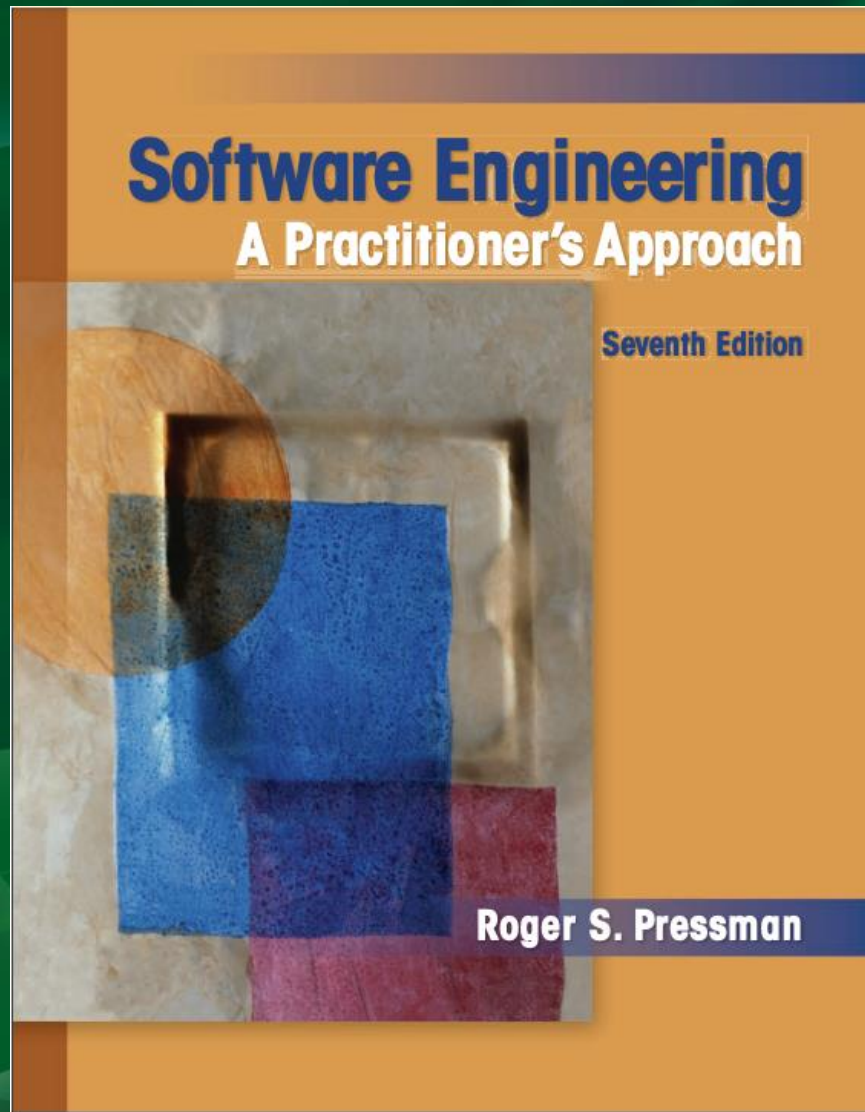
6. GRADING WEIGHTS

▸ Quizzes/Tests/Individual Assignments	5% of Total	+	⋮
▸ Proposal	5% of Total	+	⋮
▸ Sprint 1	10% of Total	+	⋮
▸ Sprint 2	10% of Total	+	⋮
▸ Design Documentation	10% of Total	+	⋮
▸ Sprint 3	15% of Total	+	⋮
▸ Sprint 4	15% of Total	+	⋮
▸ Attendance	5% of Total	+	⋮
▸ Individual Assessments	25% of Total	+	⋮

Major assignments

Assignment	Format	Due Date (Subject to change)
Organization of groups. Work with your TA to communicate who the 6 member of your team is	Canvas upload	Weeks 1 and 2
Proposal	Presentation and PPT	Weeks 3 and 4
User Stories	Spreadsheet	TBD
Poor grade report		TBD
Sprint 1 Playback	Presentation, Prototype, etc.	TBD
Sprint 2 Playback	Presentation, Code, etc.	3 weeks after Sprint 1
User Stories (iteration 3)	Spreadsheet	TBD
Peer Review (Interim)	Survey	TBD
UML Diagrams	UML tooling output	TBD
System design documentation	PDF	TBD
Sprint 3 and 4 Playback	Presentation, code, etc.	3 week cycles between Sprints
Peer Review (Final)	Survey	TBD

Recommended Textbook (on Canvas)



ATTENDANCE?



What do you have at the end of class - working software!

- Poster describing what you have created
- Presentation - "Elevator pitch"
- Demo of working application (web site, VR, AR, web, mobile, block chain, etc..)
- Source code
- A portfolio to share with prospective employers



Other Sample Projects from Last Semester



A summary of possible projects:

Cool Kids App

- Cool Kids Campaign (<https://coolkidscampaign.org/>) is an organization working with pediatric oncology patients and their families. They are interested in developing a Cool Kids Clubhouse app.
 - Login Capability
 - Display a Calendar of Events
 - Allow a user to select event(s) of interest and RSVP
 - Receive notifications related to selected events
 - Backend (which is likely to be associated with Outlook).



A summary of possible projects:

Analysis of Campus Wi-Fi Data

- We have a body of campus WIFI log data that can be investigated to develop visualizations of movement patterns over campus over a period of time.
- Something like a heat-map overlaid on a physical map of the campus area with a summary displaying characteristics would be of interest.



A summary of possible projects:

Analysis and Visualization of Fireball Data.

- NASA maintains data on meteors entering the Earth's atmosphere. The data covers the period from 1988 to the present and includes several parameters associated with each event.
- This information can be found at:
 - <https://cneos.jpl.nasa.gov/fireballs/>
 - <https://ssd-api.jpl.nasa.gov/doc/fireball.html>
- The objective of the project would be to ingest this data and develop visualizations as well as explore to extract interesting elements. For example:
 - Does the frequency of meteor events change over time?
 - Are some events connected?
 - How does the occurrence of events align with known meteor showers?
 - Can orbital debris be associated with specific events?
- Plenty of room for imagination.



A summary of possible projects:

NC Department of Insurance

- The objective is to produce a searchable database that could be used to locate code inspectors.
- This could include setting up a database, generating a Web-based interface, and - possibly - development of a mobile app.
- An official from the NC Department of Insurance will provide input and identify online data sources which can be employed.



About Your Instructor

- Started career in 1984, just as the PC revolution took off.
- Worked in S/W development lab for 6 years and realized I didn't like it.
- Switched over to a new subsidiary, IBM Consulting Group in 1994.
- Rose through the ranks to start a new brand at IBM, called IBM Interactive, where your instructor led a practice of 550 US consultants and managed and signed between \$24M - \$150M in contracts my last few years as North America Practice Executive.
- Became a Distinguished Engineer in 2005, promoted to VP in 2012.
- Retired IBM in 2018
- Had many great experiences (and a few bad clients) through the years running project teams
 - Web site consolidation
 - Marketing campaigns
 - IT Strategy
 - Web site development (Internet and Intranet)



Clients



Clients

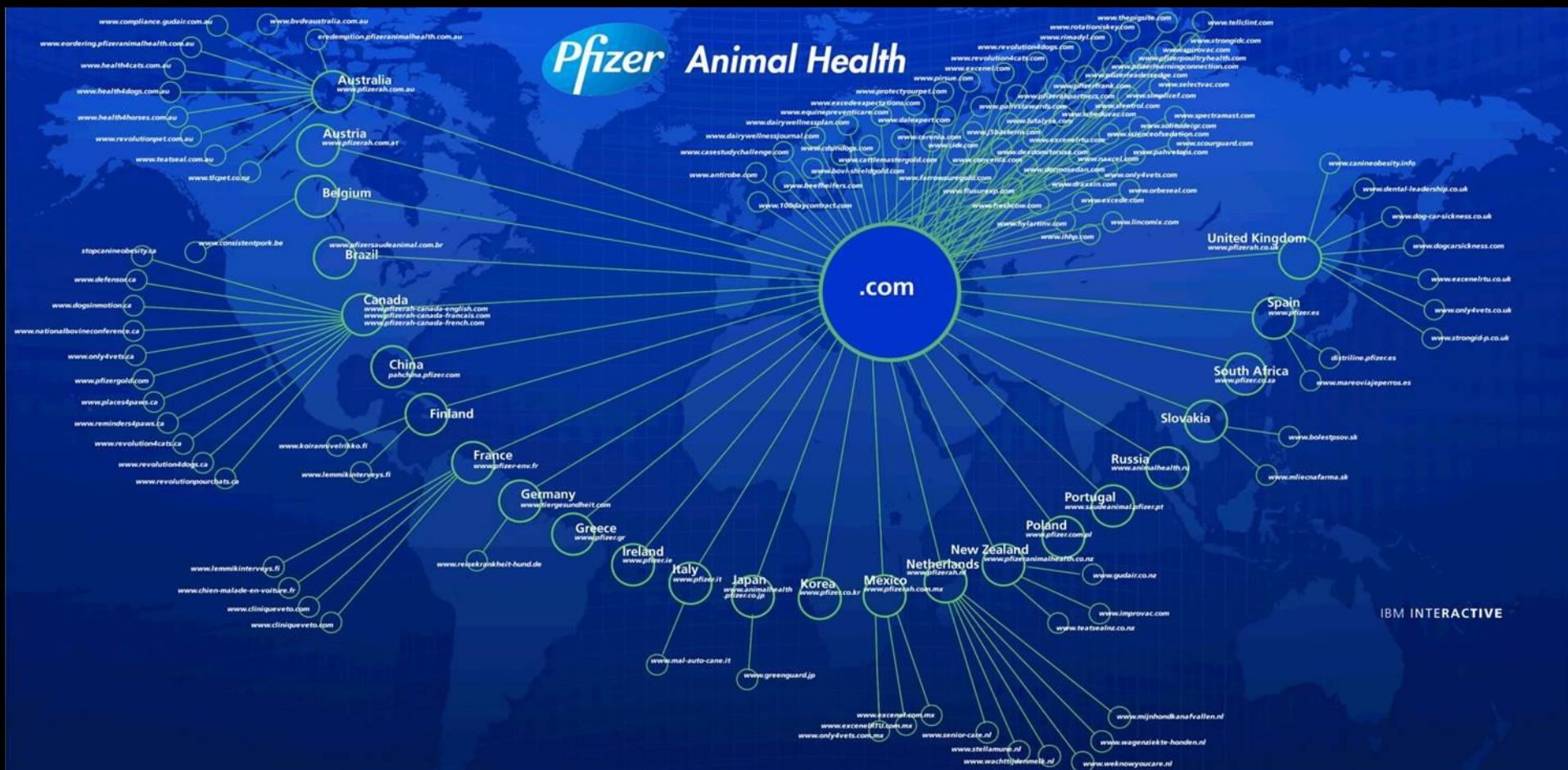


Clients



Clients





Clients



Assignment for Today

Background Survey

ⓘ This is a preview of the published version of the quiz

Started: Aug 25 at 7:51pm

Quiz Instructions

The results will help you find people to fill out your groups in case you are short on people. Some of these questions at first glance might seem redundant. They aren't.



Question 1

1 pts

What is your name?

Edit View Insert Format Tools Table

8pt ▾ Paragraph ▾ | ⋮



Assignment for next week - due next Tuesday

Proposed Groups - Only one team member per group has to do this assignment

ⓘ This is a preview of the published version of the quiz

Started: Aug 25 at 7:39pm

Quiz Instructions

As you know, 4155 is a group-based class, where there will be SIX people assigned to each group. If you have a preference for someone to be in your group (and they concur), you will be provided an opportunity to identify those group members in this survey and will get a chance to work with them during the semester.

If you pick someone to be in your group, their selection must match yours exactly. So if John Doe picks Jane Smith, and Will Jones, then Jane Smith should have John and Will in her list, and Will should have John and Jane in his list. If there is a mismatch, you will not be assigned together.

You have until next Tuesday to form your groups.

If you choose not to select your own group, you will be assigned to a group based on the answers to your background survey.

If you are in the 11:15 class, anyone else in the 11:15 class can be on your team.

If you are in the 2:30 class, anyone else in the 2:30 class can be on your team.

If you are in the 5:30 class, anyone else in the 5:30 class can be on your team.

If they are not in the same class as you, they cannot be on your team (e.g., someone from the 11:15 class can't be on a 2:30 team).



Proposed Groups (see Canvas assignment)

- Pick one team member to do this assignment
 - What is the proposed name of your group
 - Name the members of your group alphabetically (make sure to include your name):
 - The format for team members is First Name, Last Name as in Jim Rudd, not Rudd, Jim
 - List the emails of team members
 - Need 6 people for each group.
- If you are short of six members, we will supplement your team on the basis of the Background survey



Challenge yourself!

- Work to pick a project that will challenge you.
 - Learn something new
 - Look at emerging technologies. See Gartner Report for further inspiration! Be creative!
 - Necessity is the mother of invention
 - Do something you haven't done before
 - We will be evaluating your project. If too easy, we will tell you. If too hard, we will provide guidance on how you might simplify or reduce scope.



Please none of these!

- 
- Budget Manager
 - Roommate Finder
 - Hard coded data
 - Book exchange
 - No personal fitness apps (unless enabled for Google Home, Alexa, or any smart watch)



Lab Time

- Any additional questions?
- Fill out background survey.
- Find your five partners.

The screenshot shows a list of assignments in a course management system. The first assignment is 'Background Survey', due 'Aug 26 | 10 pts', with a 'Today' label and a green checkmark. The second assignment is 'Proposed Groups - Only one team member per group has to do this assignment', due 'Aug 30 |', with a green checkmark. A text box overlay at the bottom of the screenshot reads 'Form your own groups by next Tuesday'.

Assignment	Due Date	Points	Status
Background Survey	Aug 26	10 pts	Completed (Today)
Proposed Groups - Only one team member per group has to do this assignment	Aug 30		Completed

Form your own groups by next Tuesday

